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Dear Daniel,

Syama SAG pebble (scat) ore sorting — assessment and recommendation

Thank you for providing the scat assay data. We have completed the assessment you requested: updating the pebble sorting business case for Syama and testing whether STARK ore sorting is worth pursuing on your SAG scat stream. The opportunity is real but conditional, and the next spend should be on test work rather than on plant.

The work builds on a 2024 African gold mine pebble sorting study, which rejects barren scat mass, reduces the circulating load and refills the freed SAG capacity with ROM. We re-parameterised it for the converted (ex-oxide) SAG line, carrying that study's ratios and unit rates across wherever Syama site data was not available. The engine transfers unmodified; the material property that made the reference case work does not.

Principal finding

Your measured scat grade of 2.0 g/t Au is the pivotal number. The reference scats assayed 0.55 g/t against a 1.57 g/t ROM — barren material, discardable almost for free. Syama's run at 83% of ROM grade, giving a back-solved waste-to-ore contrast of only 2.23× against 8.91×. Barely a quarter of the scat stream is barren dilution — a hard ceiling on what any sorter can reject.

Table 1: Syama scats vs the 2024 African gold mine study

Parameter	Reference study	Syama
ROM / scat grade (g/t Au)	1.57 / 0.55	2.40 / 2.00
Scat grade as % of ROM	35%	83%
Heterogeneity contrast	8.91×	2.23×
Barren mass available in scats	~69%	~27%
Reject grade at 90/90 duty (g/t)	0.151	0.782

Every rejected tonne is therefore expensive. At 0.78 g/t the reject stream holds some US\$75 per tonne of recoverable gold against the US\$7.69 it avoids — nearly ten dollars of gold discarded for every dollar saved.

Economics

Your scats already recirculate and the SSCP is installing a pebble crusher for this stream, so no gold is being lost today. The case is throughput debottlenecking, not recovery, and turns on whether the freed SAG capacity can be refilled with ore.

Table 2: Outcomes at the reference 90/90 sorter duty, US\$4.1 M capital

Scenario	Net US\$/a	NPV @ 8%
Freed capacity refilled at 2.4 g/t	+9.34 M	+55.2 M
Breakeven (40% capture)	+0.63 M	0
No spare ore — ROM feed held flat	−5.17 M	−36.9 M

The upside is genuine, with a payback near five months, but the downside is equally real: a plant that cannot fill the capacity it frees destroys value and sends gold to the dump. Price is not the swing factor.

Recommendation

We recommend a conditional go: fund the test work, not the plant. Two gates should close before any capital is committed.

1. **Constraint audit** — confirm that comminution, rather than the roaster or ore supply, binds the converted line, and that ore exists at reserve grade to fill it. The published record points the other way. No cost, about four weeks.
2. **Bulk sensor test** — the STARK Phase 2 mill pebble campaign, 2 to 5 tonnes, sensor selection left open. A 2.23× mass contrast is not evidence of a detectable sensor contrast, and Syama's gold is refractory, so reject grades need fire assay. Roughly US\$60,000 to US\$150,000, about three months.

Related considerations

- We would scope coarse-ROM dilution rejection into the same campaign; cave dilution across some 2.6 Mt/a offers far more heterogeneity than the scats do.
- A sulphidic reject stream will need acid rock drainage characterisation before dumping, carrying its own closure cost.
- Given the sensitivity to reject grade, we would not skip STARK Phase 3: a scan-only analyser in circuit measures the real particle population with no gold at risk before any ejection is committed.

Reference-study figures are flagged on the model Dashboard. Its 11.9% dilution is the one we would question — STARK's own gold benchmark uses 15%. We would welcome the chance to refine these with your team.

Yours sincerely,

Mike Daniel

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Enclosures: decision brief; Syama_Pebble_Sorting_Simulation.xlsx